

Question

Elements **A**, **B**, and **C** are in the same period with increasing atomic numbers and are adjacent in the periodic table. They can all form ions with the general formula XO_m^{n-} . The value of m is the same for these three ions, and the value of n is also the same. None of these three ions can exist in large quantities in acidic aqueous solutions. It is known that the mass fraction of these three elements in XO_m^{n-} is strictly less than 50%.

Regarding these three elements, which of the following statements is correct?

- A.** Element **A** has paired valence electrons.
- B.** The AO_m^{n-} ion of element **A** is colorless.
- C.** Acidification of the BO_m^{n-} ion of element **B** releases bubbles.
- D.** Acidification of BO_m^{n-} ions of element **B** can produce a white precipitate.
- E.** The standard electrode potential of the half-reaction for the oxidation of element **C** in the CO_m^{n-} ion under acidic conditions is greater than 2.0V.
- F.** Ions of element **M** with the formula MO_m^{n-} can be used to test for phosphate ions, then element **M** belongs to the same group as element **C**.

Answer

Correct Answer: **E**

Detailed Explanation

First forming XO_m^{n-} implies that these three adjacent elements in the same period have the same valence, which basically locks it down to a transition metal, because it is difficult for three consecutive main group nonmetals to have the same valence.

CHECKPOINT

1 PTS

It is difficult for three consecutive main group nonmetals to have the same valence

To verify this hypothesis, consider the values of m and n :

When m is 1, the relative atomic mass of **X** must be less than 16, and it is impossible to have three consecutive XO^{n-} .

When m is 2, the relative atomic mass of **X** must be less than 32. At this time, because the atomic mass of sulfur is 32.06, it is impossible. The most likely candidates are **Al**, **Si**, **P**, but since n can only be 1 (metaaluminate) at this time, and there is no such species as SiO_2^- , m cannot be 2.

When m is 3, the relative atomic mass of **X** must be less than 48, and the element must be before **Ti**. Consider **P**, **S**, **Cl**. When n is 1, sulfur does not satisfy; when n is 2, chlorine does not satisfy; when n is 3, sulfur does not satisfy; when it is greater than 3, phosphorus does not satisfy. Therefore, m does not take 3 either.

CHECKPOINT

2 PTS

m cannot take values 1, 2, 3

When m is 4, the relative atomic mass of **X** must be less than 64, and the most likely candidates are transition metals before **Cu**. Consider n being 1, but MnO_4^- is stable to acid, which does not meet the requirements, so n takes 2.

CHECKPOINT

1 PTS

MnO_4^- is stable to acid, which does not meet the requirements, so n takes 2

It is easy to associate with chromate ion CrO_4^{2-} , which is unstable under acidic conditions and transforms into dichromate ion; the adjacent element can also form manganate ion MnO_4^{2-} , which is unstable under acidic conditions and undergoes self-disproportionation; the element adjacent to it is iron, and FeO_4^{2-} is ferrate ion, which is unstable under acidic conditions and can oxidize water. Therefore, the elements **Cr**, **Mn**, **Fe** meet the requirements of the question. **A** is **Cr**, **B** is **Mn**, **C** is **Fe**.

CHECKPOINT

1 PTS

CrO_4^{2-} , MnO_4^{2-} , FeO_4^{2-} are all unstable to acid

CHECKPOINT

2 PTS

A is **Cr**, **B** is **Mn**, **C** is **Fe**.

Therefore, we begin to judge the options:

A. **Cr** has a half-filled orbital and does not have paired valence electrons, A is incorrect.

CHECKPOINT

1 PTS

Cr has a half-filled orbital and does not have paired valence electrons, A is incorrect.

B. CrO_4^{2-} appears yellow, B is incorrect.

CHECKPOINT

1 PTS

CrO_4^{2-} appears yellow, B is incorrect.

C. The disproportionation reaction of MnO_4^{2-} under acidic conditions is



CHECKPOINT

1 PTS



Disproportionation does not produce gas, C is incorrect.

CHECKPOINT

0.5 PTS

Disproportionation does not produce gas, C is incorrect.

D. The disproportionation of MnO_4^{2-} produces a brownish-black manganese dioxide precipitate, not white, D is incorrect.

CHECKPOINT

0.5 PTS

The disproportionation of MnO_4^{2-} produces a brownish-black manganese dioxide precipitate, not white, D is incorrect.

E. FeO_4^{2-} ion has strong oxidizing properties, and the standard electrode potential of $\text{FeO}_4^{2-}/\text{Fe}^{3+}$ under acidic conditions is $2.2V$, E is correct.

CHECKPOINT

1 PTS

The standard electrode potential of $\text{FeO}_4^{2-}/\text{Fe}^{3+}$ under acidic conditions is $2.2V$, E is correct

F. The MO_4^{2-} ion used to test for phosphate is molybdate ion MoO_4^{2-} ion, which belongs to the same group as **Cr** and not **Fe**, F is incorrect.

CHECKPOINT

1 PTS

The MO_4^{2-} ion used to test for phosphate is molybdate ion MoO_4^{2-} ion, which belongs to the same group as **Cr** and not **Fe**, F is incorrect

In summary, option E is correct.